

**Coverage Policy Manual**

Policy #: 1998076
Category: Surgery
Initiated: February 1998
Last Review: May 2025

Meniscal Allograft Transplantation and Synthetic Meniscal Implants**Description:**

Meniscal cartilage is an integral structural component of the human knee, functioning to absorb shocks and providing load sharing, joint stability, congruity, proprioception, and lubrication and nutrition of the cartilage surfaces. Total and partial meniscectomy frequently result in degenerative osteoarthritis. The integrity of the menisci is particularly important in knees in which the anterior cruciate ligament has been damaged. In these situations, the menisci act as secondary stabilizers of anteroposterior and varus-valgus translation.

Meniscal allograft transplantation (MAT) is considered a salvage procedure, reserved for patients with disabling knee pain following meniscectomy who are considered too young to undergo total knee arthroplasty or in patients who require a total or near total meniscectomy for irreparable tears. As a result, the population intended to receive these transplants is relatively limited. Using a large database of privately insured non-Medicare patients, Cvetanovich et al estimated an annual incidence of MAT in the U.S. of 0.24 per 100,000 (Cvetanovich, 2015). It is not expected that clinical trials will be conducted to compare meniscal allografts with other orthopedic procedures, although trials comparing allograft transplant with medical therapy are possible.

There are 3 general groups of patients who have been treated with MAT:

- young patients with a history of meniscectomy who have symptoms of pain and discomfort associated with early osteoarthritis that is localized to the meniscus-deficient compartment
- patients undergoing anterior cruciate ligament reconstruction in whom a concomitant meniscal transplant is intended to provide increased stability
- young athletes with few symptoms in whom the allograft transplantation is intended to deter the development of osteoarthritis. Due to the risks associated with this surgical procedure, prophylactic treatment for this purpose is not frequently recommended

Issues under study include techniques for processing and storing the grafts, proper sizing of the grafts, and the most appropriate surgical techniques. The four primary ways of processing and storing allografts are fresh, fresh frozen, cryopreserved, and lyophilized. Fresh viable implants, harvested under sterile conditions, are less frequently used since the grafts must be used within a couple of days to maintain viability. Alternatively, the harvested meniscus can be fresh frozen for storage until needed. Cryopreservation freezes the graft in glycerol, which aids in preserving the cell membrane integrity and donor fibrochondrocyte viability. Cryolife is a commercial supplier of such grafts. Donor tissues may also be dehydrated (freeze-dried or lyophilized), permitting storage at room temperature. Lyophilized grafts are prone to reduced tensile strength, shrinkage, poor rehydration, post-transplantation joint effusion, and synovitis; these are no longer used in the clinical setting. Several secondary sterilization techniques may be used, with gamma irradiation the most common. The dose of radiation considered effective has been shown to change the mechanical structure of the allograft; therefore, non-irradiated grafts from screened donors are most frequently used. In a survey conducted by the International Meniscus Reconstruction Experts Forum, when surgeons were asked about allograft preference, 68% preferred fresh frozen nonirradiated allografts, with 14% responding fresh viable allografts (Getgood, 2017).

There are several techniques for MAT; most are arthroscopically assisted or all-arthroscopic. Broadly, the techniques are either all-suture fixation or bone fixation. Within the bone fixation category, the surgeon may use either bone plugs or a bone bridge. Types of bone bridges include keyhole, trough, dove-tail, and bridge-in-slot. The technique used depends on laterality and the need for concomitant procedures. Patients with malalignment, focal chondral defects, and/or ligamentous insufficiency may need concomitant procedures (osteotomy, cartilage restoration, and/or ligament reconstruction, respectively) (Frank, 2015).

Tissue engineering that grows new replacement host tissue for individual patients is also being investigated. For example, the Collagen Meniscus Implant (CMI) (by Stryker, formerly the ReGen Collagen Scaffold by ReGen Biologics), is a resorbable collagen matrix comprised primarily of type I collagen from bovine Achilles tendons. The implant is provided in a semilunar shape and trimmed to size for suturing to the remaining meniscal rim. The implant provides an absorbable collagen scaffold that is replaced by the patient's soft tissue; it is not intended to replace normal body structure. Because it requires a meniscal rim for attachment, it is intended to fill meniscus defects after a partial meniscectomy. A second collagen meniscus implant, RejuvaKnee has similar characteristics to CMI; however, the bovine collagen is sourced from the meniscus as opposed to the Achilles tendon. Other scaffold materials and cell-seeding techniques are being investigated. Non-absorbable and non-porous synthetic implants for total meniscus replacement are in development. One total meniscus replacement that is in early phase clinical testing is NUSurface (Active Implants), which is composed of a polyethylene reinforced polycarbonate urethane.

The outcomes of this treatment (i.e., pain, functional status) are subjective, patient-reported outcomes that are prone to placebo effects. On the other hand, the natural history of a severely damaged meniscus is predictable, with progressive joint damage, pain, and loss of function.

Regulatory Status

In 2008, the ReGen Collagen Scaffold was cleared for marketing by the U.S. Food and Drug Administration (FDA) through the 510(k) process. The FDA determined that this device was substantially equivalent to existing absorbable surgical mesh devices. The ReGen Collagen Scaffold (also known as MenaFlex CMI) was the only collagen meniscus implant with FDA clearance at that time. Amid controversy about this 510(k) clearance, the FDA reviewed its decision. In October 2010, the FDA rescinded the approval, stating that MenaFlex is intended for different purposes and is technologically dissimilar from the predicate devices identified in the approval process. The manufacturer appealed the rescission and won its appeal in 2014. The product, now called CMI, was manufactured by Ivy Sports Medicine (now Stryker). A second collagen meniscus implant, RejuvaKnee (Collagen Matrix, Inc [now Regenity]), was declared substantially equivalent to CMI by the FDA in 2024.

FDA product code: OLC.

Coding

Effective in 2005, there is a CPT category I code specific to this procedure when performed arthroscopically:

29868: Arthroscopy, knee, surgical; meniscal transplantation (includes arthrotomy for meniscal insertion), medial or lateral

There is no CPT code for implantation of the ReGen Collagen Scaffold, but the American Academy of Orthopaedic Surgeons' Coding, Coverage and Reimbursement Committee feels that the meniscal transplantation CPT code 29868 is appropriate for this procedure.

**Policy/
Coverage:****Effective, March 2011****Meets Primary Coverage Criteria Or Is Covered For Contracts Without Primary Coverage Criteria**

Meniscal allograft transplantation for patients with persistent or recurrent knee pain despite prior meniscectomy or partial meniscectomy meets member benefit certificate primary coverage criteria that there be scientific evidence of effectiveness for patients who meet ALL of the following criteria:

- Persistent activity related knee pain (lasting at least 6 months) that is refractory to physical therapy and analgesic medications
- Adult patients should be too young to be considered an appropriate candidate for total knee arthroplasty or other reconstructive knee surgery (e.g., younger than 55 years)
- Absence or near absence (>50%) of the meniscus, established by imaging or prior surgery
- Knee joint with stable ligaments and normal alignment
- Documented minimal to absent degenerative changes in the surrounding articular cartilage

Meniscal allograft transplantation when performed in combination, either concurrently or sequentially, with autologous chondrocyte implantation or osteochondral allografting for focal articular cartilage lesions meets member benefit certificate primary coverage criteria that there be scientific evidence of effectiveness in improving health outcomes.

Does Not Meet Primary Coverage Criteria Or Is Investigational For Contracts Without Primary Coverage Criteria

Synthetic (e.g., collagen or polyurethane) meniscus implants do not meet member benefit certificate primary coverage criteria that there be scientific evidence of effectiveness.

For members with contracts without primary coverage criteria, synthetic (e.g., collagen or polyurethane) meniscus implants are considered investigational. Investigational services are specific contract exclusions in most member benefit certificates of coverage.

Effective October 2009 – February 2011

Meniscal allograft transplantation for patients with persistent or recurrent knee pain despite prior meniscectomy or partial meniscectomy meets member certificate of benefit Primary Coverage Criteria for effectiveness for patients with:

- Persistent activity related knee pain (lasting at least 6 months) that is refractory to physical therapy and analgesic medications
- Between 15 and 55 years old (adolescents should be skeletally mature with documented closure of growth plates)
- Absence or near absence (>50%) of the meniscus, established by imaging or prior surgery
- Knee joint with stable ligaments and normal alignment
- Documented minimal to absent degenerative changes in the surrounding articular cartilage

This coverage is based on studies of meniscal transplantation which have short to medium outcomes. Long-term outcomes of meniscal transplantation have not been reported. The potential for transmission of infection with known viruses or other unknown infectious agents remains a concern in relation to this procedure.

Meniscal allograft transplantation when performed in combination, either concurrently or sequentially, with autologous chondrocyte implantation or osteochondral allografting or for any other indication does not meet primary coverage criteria of effectiveness.

Collagen meniscus implants do not meet member benefit certificate Primary Coverage Criteria that there be scientific evidence of effectiveness in improving health outcomes.

For contracts without primary coverage criteria, meniscal allograft transplantation is considered investigational when performed in combination, either concurrently or sequentially, with autologous chondrocyte implantation or osteochondral allografting or for any other indication.

For contracts without Primary Coverage Criteria, collagen meniscus implants are considered investigational. Investigational services are an exclusion in the member benefit certificate of coverage.

Effective, June 2009

Meniscal allograft transplantation for patients with persistent or recurrent knee pain despite prior meniscectomy or partial meniscectomy meets member certificate of benefit Primary Coverage Criteria for effectiveness for patients with:

- Persistent activity related knee pain (lasting at least 6 months) that is refractory to physical therapy and analgesic medications
- Between 15 and 55 years old (adolescents should be skeletally mature with documented closure of growth plates)
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For contracts without primary coverage criteria, meniscal allograft transplantation is considered investigational when performed in combination, either concurrently or sequentially, with autologous chondrocyte implantation or osteochondral allografting or for any other indication.

Effective, February 1998

Meniscal allograft is considered investigational and is not covered.

Rationale:

Intermediate outcomes regarding meniscal allograft transplantation primarily focus on the viability of the transplanted tissue. Long-term outcomes vary with the patient population studied. For example, relief of pain and improved function are critical outcomes for symptomatic patients with signs and symptoms of osteoarthritis. When performed in conjunction with repair of the anterior cruciate ligament (ACL), improved function and maintenance of knee stability is pertinent. In both of the above situations, it is important to isolate the contribution of the meniscal allograft to the overall surgical procedure. Finally, for asymptomatic patients who are undergoing allograft transplantation prophylactically, long-term outcomes regarding the incidence of subsequent osteoarthritis is important.

Data regarding meniscal allograft transplantation are of poor quality. For example, none of the studies reporting health outcomes included preoperative and postoperative measures of restoration of knee function, including MRI results or second-look arthroscopy. None of the studies presented clear comparisons of preoperative clinical findings to postoperative results. Each study assessed outcomes differently. While definitive data were not available, in general, poor results were reported in patients with Outerbridge grade III or IV osteoarthritis, or in those with unstable knees, and thus researchers have largely abandoned meniscal allograft transplantation in these patients.

The literature published since the 1997 Assessment does not address the limitations identified in the Blue Cross Blue Shield Association Technology Evaluation Center Assessment. In terms of the intermediate outcome of graft viability, the largest case series has been collected by CryoLife, a commercial supplier of cryopreserved allografts. However, these data are not available in the published peer-reviewed literature. As summarized by Johnson in a 1999 report, among 1,023 transplants, CryoLife reported graft survival of 93% when the meniscus is transplanted with a bone plug for fixation, compared to 67% without such fixation. The method of determining graft viability, with either serial MRI scans or second-look arthroscopy, is not reported. Additional patient outcome information is posted on their company Web site, based on responses to patient questionnaires. A total of 332 patients were contacted; 136 (41%) responded. The mean patient age was 35 years old, and in 84% of cases transplantation was related to prior sports or traumatic injury. A total of 80% of patients rated their knee function as normal to nearly normal as compared to their knee function before surgery. The incomplete nature of the data and the heterogeneous population of patients, many of whom presumably underwent concomitant knee reconstructive procedures, make this data scientifically uninterpretable.

2003 Update

A review of the peer-reviewed literature on MEDLINE for the period of 2001 through February 2003 found no articles addressing the study limitations noted above. In a case study of 23 patients, Rath and colleagues reported significant improvement in function and reduced pain as measured by the Short Form-36 scores after cryopreserved meniscal allograft transplantation for compartmental pain after total meniscectomy 2-8 years post-operatively. However, the authors noted function remained limited, and 8 of 22 allografts tore during the study period and required

total or partial meniscectomies. In a prospective study of 23 patients who underwent medial meniscal transplantation with reconstruction of the ACL, Wirth and colleagues reported better results in the 6 cases of preservation of deep-frozen allografts over lyophilized meniscal transplants at 3 and 14 years postoperatively. While the Wirth study used 2 ACL control groups for comparison, the concomitant knee reconstruction does not permit scientific analysis attributable to meniscal allograft transplantation alone. In addition, no preoperative clinical outcomes were assessed. Therefore, these 2 studies do not alter the findings above and the policy statement is unchanged.

March 2007 Update

A paper on "current concepts" on Meniscus allograft transplantation by Lubowitz, PC Verdonk, JB Reid III, and R Verdonk published in May 2007 was reviewed. The authors stated, "A growing body of evidence suggests that pain relief and functional improvement may reliably be achieved at short- and medium-term follow-up, and even, in some cases, at long term (>10 years) follow-up. Future research must address the issue of optimal timing of the procedure and whether meniscal transplantation results in demonstrable long-term benefits, especially with regard to protection of articular cartilage."

December 2007 Update

Two articles by Sekiya et al and Rankin et al were reviewed. All articles reported retrospective studies with small numbers of patients and short follow-up periods. Sekiya et al, in 2003, concluded a prospective randomized controlled study would be necessary to definitively prove the impression that restoration of meniscal function would provide protection for the articular cartilage and improve joint stability.

R Verdonk, KF Almqvist, W Huyesse, and PC Verdonk published another review in September 2007, stating, "...the major drawback in all meniscus allograft studies is the general lack of a control population."

Another September 2007 article by G von Lewinski, et al., reporting their 20 year experience with meniscal allograft transplantation described that radiological results revealed clear degenerative changes with long-term follow-up after meniscal allograft transplantation even though some patients did relatively well regarding the subjective and clinical results...in comparison with the literature.

2009 Update

In 2007, Matava published a systematic review of the available literature; none of the 15 studies identified could be classified as Level I or Level II (prospective controlled comparisons), 3 studies qualified as Level III (retrospective comparisons); the 12 remaining studies were retrospective case series. The primary indication for meniscal allograft transplantation in these studies was complete or near-complete meniscectomy with pain in the involved compartment, and before the development of moderate to severe arthrosis (less than 2 to 3 mm of joint space narrowing and/or limited chondral wear) in a young (less than 50 years of age) active patient. Lower extremity malalignment and/or ligamentous instability have been associated with meniscal transplantation failure, and thus were treated (e.g., osteotomy or ACL repair) before or at the time of the transplantation. Twelve studies used validated outcome measures, with second-look arthroscopy conducted in some of the patients in 11 studies. "Success" rates were usually over 60% (ranging from 13% to 100%), with more recent series reporting short-term favorable outcomes (based on pain, function, and patient satisfaction) in about 85% of their patient cohorts (generally 20 to 30 patients per cohort). Up to 26% reoperation rates for allograft tears in addition to other complications were reported. Matava et al noted that fresh frozen or cryopreserved grafts were associated with the highest success rates and the least risk for biomechanical degradation or disease transmission.

Also, in 2007, Hommen and colleagues reported 10-year follow-up on 20 of 22 consecutive patients who received cryopreserved meniscus allografts. The 10-year graft survival/success rate was 45%, with 5 allograft failures identified on second-look surgery, 5 allografts with grade III tears identified on magnetic resonance imaging (MRI), and 4 patients reporting no improvement. Of 15 patients with follow-up radiographs, 10 had narrowing and 12 had progression of the Fairbank degenerative joint disease score in the transplanted tibiofemoral compartment. Twenty-year follow-up was reported for 5 patients who had received a deep frozen meniscal allograft along with other procedures on the knee (Lewinski et al, 2007). At 20-year follow-up MRI revealed shrinkage of the transplants with very small rims of the meniscus; the remaining meniscal tissue showed degenerative changes. As noted by Hommen et al, questions remain about whether meniscus grafts can delay or prevent the progression of degenerative changes and joint space narrowing (Hommen et al, 2007).

In summary, meniscal allograft transplantation appears to improve symptoms in some patients with a prior meniscectomy who are considered too young to undergo total knee replacement.

Evidence consisting primarily of retrospective case series indicates that this procedure may produce short- term to intermediate-term pain relief in selected patients. At this time, there are no long-term controlled studies. The literature does not permit conclusions concerning the effect of meniscal transplantation on the progression of degenerative changes and joint space narrowing.

Meniscal allograft transplantation is associated with a high number of complications, including tears of the transplanted meniscus, displacement, or arthrofibrosis. In addition, the risk of

infection from cadaver transplant has not been completely resolved. Careful selection of patients and surgical technique appear to be critical for success of this procedure.

Collagen Meniscus Implant

In 2006, a German research group published initial results from a randomized trial that compared high tibial valgus osteotomy alone with osteotomy plus a collagen meniscus implant in 60 patients with subtotal loss of the medial meniscus (Linke, 2006). Arthroscopy on the first 23 of 30 patients with a collagen implant at 8-18 months post-surgery showed complete healing in 8 patients (35%), partial healing in 7 (30%; requiring resection of the posterior part of the implant), and poor results with only small remains of the collagen implant left in 8 patients (35%). Complications included implantation in insufficiently vascularized tissue, sutures cutting into the implant, inadequate fixation to the rim, destruction of the implant in an unstable knee joint or with premature loading post-operatively, allergic reaction to the xenogenic collagen implant, avulsion of the implant with joint blocking, and infection. Assessment of pain and function (Lysholm, International Knee Documentation Committee, subjective pain scores) showed slight and non-significant differences in comparison with the 16 patients treated with correction osteotomy only. Longer follow-up on all 60 patients is reported to be continuing.

Data provided to the FDA in support of the 510(k) application for the Menaflex collagen meniscus implant included a prospective, randomized multicenter clinical trial (26 surgeon-investigators from 16 sites) that was conducted under an investigational device exemption (IDE) and published in 2008 (Rodkey, 2008). The study involved 311 patients, randomized and analyzed separately for those with no prior surgery (acute group, n = 157) and those who had from 1 to 3 prior meniscal procedures (chronic group, n = 154). Included were patients 18 to 60 years of age who had an irreparable injury to or previous partial loss of one medial meniscus, with an intact rim. The involved knees had to be in neutral alignment, and patients with a fullthickness (Outerbridge Grade-IV) chondral lesion were excluded from the study, as were patients with posterior cruciate ligament insufficiency. Patients within the acute and chronic arms were randomized to receive the collagen meniscus implant or be treated with a partial meniscectomy only. Control patients (partial meniscectomy only) underwent standard physical therapy only. Patients receiving a collagen meniscus implant wore a knee brace that was locked in full extension for 6 weeks and were required to be non-weight-bearing with crutches for 2 weeks. The brace was removed 3 to 4 times per day to perform self-assisted passive range-of-motion exercises. Full weight-bearing and unlimited active and passive range-of-motion exercises were initiated after 6 weeks. Second-look arthroscopy with biopsy at one year was performed for all patients who received a collagen meniscus implant (acute and chronic); 141 of 160 patients in the 2 groups (88%) had arthroscopy. Patients underwent frequent clinical follow-up examinations over 2 years and completed validated outcomes questionnaires for up to 7 years (Lysholm functional score, Tegner activity scale and pain on a visual analog scale). At an average follow-up of 59 months (range, 16-92), new tissue was reported to appear grossly meniscus-like and integrate with the meniscus rim, with about 50% of the defect filled (45% in the acute group and 58% in the chronic group). There were no differences between the implant and control groups for pain, Lysholm and self-assessment scores. Only Tegner activity scores in the chronic arm were significantly different, with patients in the implant group reporting regaining 42% of their lost activity level, compared to a 29% regain in activity reported by controls. Implant patients in the chronic group underwent significantly fewer non-protocol reoperations (8 vs. 15) compared to controls, although the report did not indicate whether procedures other than a "re-look" were performed during the one-year protocol scheduled reoperation. No differences were detected between the 2 treatment groups in the acute arm of the study, with 5 reoperations in each group. Kaplan-Meier analysis, which was estimated at 5 years due to the low number of patients at risk, suggested a modest increase in survival in the chronic group who had received a collagen implant compared to controls (about 90% vs. 80%). Longer follow-up is needed to determine whether implantation of a collagen scaffold is able to slow joint degeneration, reduce pain, or otherwise improve the net health outcome at longer durations. Controlled studies comparing the collagen meniscus implant with allograft transplantation would also be of value in evaluating the health benefit of this procedure. The evidence at this time is insufficient to permit conclusions concerning the effect of this technology on health outcomes.

2011 Update

Meniscal Allograft Transplantation

In 2010 Vundelinckx et al reported 5 to 15 year follow-up from 49 patients (69%) who had received meniscal allograft transplantation (Vundelinckx, 2010). Five of the patients were considered failures (10%) as they underwent TKA because of persistent pain, 8 patients were contacted but chose not to join the study, and 2 patients were lost to follow-up. The 34 patients who participated in the follow-up study had a mean age of 33 years (range 14, to 47 years) at the time of transplantation. At an average 105-month follow-up there was a significant decrease in the VAS (7 to 3.4) and increase in the Knee Injury and Osteoarthritis Outcome Score (KOOS, 35.8 to 60.2), Lysholm (39.7 to 71.8) and total Short Form (SF)-36 Health Survey (51.5 to 75.2) relative to preoperative levels. There was no increase in Tegner activity level. The more severe the osteoarthritis at follow-up the less the improvement in the KOOS and Lysholm scores. Radiographic evaluation showed an overall increase in osteoarthritis compared to baseline (0.58 points on the Kellgren-Lawrence classification), with a slight or moderate increase in osteoarthritis in 42% of the patients (1 or 2 points), and no increase in osteoarthritis in 58% of patients. No correlations were found between outcome scores and preoperative cartilage damage (no more than grade 3 Outerbridge was an exclusion), preoperative osteoarthritis, alignment deviation, gender, or body mass index. Kaplan-Meier analysis showed that a majority of the failures occurred about 10 years post-operatively.

The 2011 literature update identified 2 systematic reviews on meniscal allograft transplantation (Elattar, 2011) (Hergan, 2011). ElAttar et al. included 44 case series and cohort studies with 1,136 grafts in 1,068 patients (mean age of 34.8 years) (Elattar, 2011). The follow-up period ranged from 8 month to 20 years, with an overall average of 4.6 years.

Twelve different clinical scoring systems were described; the Lysholm and the International Knee Documentation Committee (IKDC) were the most commonly used. The average overall Lysholm score increased from 44 at baseline to 77 at the latest follow-up, the Tegner activity score increased from 3 to 5, while the mean VAS for pain decreased from 4.8 to 1.7. Most of the studies found only slight or no loss of the joint space in a majority of patients, while some studies reported an increase in joint space in some patients. Magnetic resonance imaging (MRI) indicated frequent good healing and incorporation of the graft, although tears, shrinkage and/or extrusion were not uncommon. With failure defined as (sub)total destruction/removal of the graft, the overall mean failure rate per trial was 10.6%. The overall complication rate was 21.3%. Hergen and colleagues conducted a systematic review of the literature to evaluate characteristics of patients, graft survival and clinical outcomes (Hergen, 2011). Fourteen English language studies with a minimum of 2 years' follow-up were included; all but one provided Level IV evidence (case series). A total of 196 knees in 9 studies were assessed for joint space narrowing after meniscal allograft transplantation. Patients with Outerbridge scores of 2 or less in any area had significantly improved post-treatment Lysholm and Tegner scores, whereas patients with Outerbridge grade 3 or greater in any area (not repaired) did not have significant improvements in post-treatment Lysholm and Tegner scores. Reported failures of the meniscal allografts at 2 years or earlier ranged from 7% to 35%. Studies that analyzed patients undergoing concomitant procedures did not detect a difference between the subgroup in comparison with meniscal allograft transplantation alone. Functional outcomes were considered generally good where reported.

Combined Meniscus Transplantation and Articular Cartilage Repair

Harris et al. published a systematic review of combined meniscal allograft transplantation and cartilage repair/restoration in 2010 (Harris, 2011). Six level IV studies (case series) with a total of 110 patients were included in the review. Patients underwent meniscal allograft transplantation with either autologous chondrocyte implantation (ACI, n=73), osteochondral allograft (n=20), osteochondral autograft (n=17) or microfracture (n=3). All studies showed improvement in clinical outcomes at final follow-up compared to the preoperative condition. Outcomes were also compared with historical outcomes, extracted from midterm and long-term follow-up studies, of individual procedures performed in isolation. Four of the 6 studies found outcomes equivalent to procedures performed in isolation, while 2 studies found that outcomes with combined surgery were not as good as the historical controls. Across the 6 studies, 13 failures (12%) were reported; these included 11 isolated meniscal allograft transplantation failures, 1 combined meniscal allograft and ACI failure, and 1 isolated ACI failure. Three knees with failed meniscal allograft transplantation were converted to total knee arthroplasty. Nearly 50% of the patients underwent 1 or more subsequent surgeries after combined meniscal allograft transplantation and cartilage repair/restoration procedures.

The largest and longest study to report on meniscal allograft transplantation in patients with significant (grade III and IV) chondral damage is by Stone et al in 2010, who described combined meniscal allograft transplantation with focal articular cartilage repair in 115 consecutive patients (119 transplants), with a mean follow-up of 5.8 years (range 2 months to 12.3 years). Fifty-three (46%) of the patients were over the age of 50 at the time of surgery, and the mean time from injury to surgery was 14.2 years (2 months to 39.7 years). The intraoperative Outerbridge classification was grade III in 22 knees (18.5%) and grade IV in 97 knees (81.5%). There was a mean of 5 concomitant procedures performed, which included articular cartilage paste grafting (n = 67) and microfracture (n=69). Thirteen patients (11%) were lost to follow-up. A total of 56 knees (47%) required up to 5 subsequent operations after the original meniscal allograft transplantation. Subsequent procedures included partial meniscectomy in 23 knees (19.3%), repair of the allograft in 11 knees (9.2%), and revision meniscal allograft transplantation in 8 knees (6.7%). Fourteen of these 42 revisions (33%) ultimately failed. Out of the 119 meniscal allograft transplantations, 25 (20.1%) failed, with 18 progressing to joint replacement. The mean survival of the allograft was 9.9 years (range 2 months to 12.3 years).

Based on these studies and expert opinion, the coverage statement is being revised to include meniscal allograft transplantation when performed in combination with autologous chondrocyte implantation, osteochondral allografting or osteochondral autografting for focal articular cartilage lesions.

Collagen Meniscus Implants

Literature updates using the MEDLINE® database through February 2011 identified two case series from Europe with the collagen meniscus implant (Bulgheroni, 2010) (Zaffagnini, 2011). Bulgheroni et al. reported 2-5 year follow-up on 28 patients who had received an implant for irreparable meniscus lesions. (30) Zaffagnini et al. compared minimum 10-year follow-up from 18 patients who chose to receive a collagen meniscus implant with 18 patients who chose a partial medial meniscectomy during the same time period (Zaffagnini, 2011). Reconstruction of the anterior cruciate ligament (ACL) and focal cartilage repair (microfracture) procedures were performed as needed. The 2 groups (non-consecutive patients meeting the inclusion and exclusion criteria) were comparable at baseline for this prospective cohort study. The control group underwent standard physical therapy while patients receiving a collagen meniscus implant wore a knee brace that was locked in full extension for 6 weeks and were required to be non-weight-bearing with crutches for 2 weeks. The brace was removed 4 times per day to allow continuous passive motion and cycling exercise. All patients followed a rehabilitation protocol for 6 months until they returned to full unrestricted activity as tolerated. At a mean of 133 months after surgery (range, 120 to 152 months), 33 patients (92%) were available for follow-up. The implanted group showed a lower VAS for pain (1.2 vs. 3.3) and higher objective IKDC score (7A and 10B vs. 4B and 12C), Tegner index (7.5 vs. 5.0) and Short Form 36 Health Survey (53.9 vs. 44.1). No significant differences were found in the Lysholm and Yulish scores. Independent and blinded radiographic evaluation showed significantly less medial joint space narrowing in the implant group than in the partial meniscectomy group (0.48 vs. 2.13 mm). MRI of the implanted knees showed 11 cases of myxoid degeneration signal, 4 knees with a normal signal but reduced size, and 2 knees that had no recognizable implant. Two patients in each group (12%) required additional surgeries. Randomized controlled trials on a larger population are necessary to confirm these results.

The American Academy of Orthopaedic Surgeons (AAOS) stated in 2009 that a meniscal transplant may be recommended for active people younger than 55, with the goal of replacing the meniscus cushion before the articular cartilage is damaged. The hope is that the transplant will also delay the development of arthritis, but long-term results are not yet available.

2012 Update

A search of the MEDLINE database was conducted through September 2012. There was no new information identified that would prompt a change in the coverage statement.

A systematic review of the Menaflex collagen meniscus implant was published in 2012 (Harston, 2012). Included were 11 studies with a total of 520 subjects, 321 of whom received a collagen meniscus implant. Research quality was generally rated as low, with widely ranging scores. Of all the patients who received a CMI, 41.1% had concomitant procedures such as anterior cruciate ligament (ACL) reconstruction or repair (n=122), high tibial osteotomy (n=33), microfracture or the femoral condyle or patella (n=5), or autologous chondrocyte implantation (n=19). The last follow-up was at a mean of 46.6 months (range, 6-135 months). Meniscus-like fibro-cartilaginous tissue ingrowth was observed, although the collagen meniscus implant and regenerated tissue were smaller than the original implant.

Approximately 66%-70% of patients who received a collagen meniscus implant had satisfactory outcomes, but in studies that had either control or comparison groups, outcomes improved in both groups. These results reinforce the need for controlled trials.

2013 Update

A literature search was conducted using the MEDLINE database through March 2013. There was no new literature identified that would prompt a change in the coverage statement.

The Actifit Study Group reported on 1-year tissue ingrowth and 2-year clinical outcomes from a prospective multi-center series of 52 patients who received a polyurethane scaffold at the time of partial meniscectomy (Verdonk, 2011; Verdonk, 2012). MRI at 3 months showed evidence of tissue ingrowth in 81.4% of patients. Out of 44 second-look arthroscopies at 1 year, 43 (97.7%) showed integration of the scaffold with the native meniscus. Biopsy specimens taken at this time showed vital material with no signs of cell death, necrosis, or adverse reaction to the scaffold material. Nine treatment failures (17.3%) occurred during the study and 5 adverse events were considered to be related to the scaffold. Two-year clinical follow-up in 39 patients (75%) found significant improvements from the pre-surgery baseline in VAS for pain, IKDC, Lysholm, and Knee Injury and Osteoarthritis Outcome Score (KOOS). Interpretation of these results is limited by the absence of a control group undergoing partial meniscectomy without the scaffold. However, the progressive improvement in clinical outcomes observed over the 2-years of follow-up does provide indirect support for successful integration of the scaffold.

Controlled trials are needed to determine the efficacy of the polyurethane meniscal scaffold with greater certainty. It is also noted that the Actifit meniscal scaffold is not currently approved for marketing in the U.S.

Practice Guidelines

The 2012 guidance from the United Kingdom's National Institute for Health and Clinical Excellence (NICE) stated that evidence on partial replacement of the meniscus of the knee using a biodegradable scaffold raised no major safety concerns, but evidence for any advantage of the procedure over standard surgery was limited (NICE, 2012). Therefore, NICE recommends that this procedure should only be used with special arrangements for clinical governance, consent and audit or research.

2014 Update

A literature search conducted through March 2014 did not reveal any new information that would prompt a change in the coverage statement. The key identified literature is summarized below.

A 2014 report from the Actifit Study group evaluated the Actifit biodegradable polyurethane scaffold for the lateral meniscus in 54 patients with postmeniscectomy syndrome (Bouyarmine, 2014). It is possible that the patient population in this study overlapped with the population in the study by the Actifit Study Group previously described (Verdonk, 2011; Verdonk, 2012). Using last observation carried forward for missing data, VAS and IKDC improved over the course of the study. At 2-year follow-up, VAS had decreased from 5.5 to 2.9, IKDC improved from 47.0 to 67.0, and most of the KOOS subscores were significantly improved.

Controlled trials are needed to determine the efficacy of the polyurethane meniscal scaffold with greater certainty. It is also noted that the Actifit meniscal scaffold is not currently approved for marketing in the U.S.

Practice Guidelines and Position Statements

The 2012 guidance from the United Kingdom's National Institute for Health and Clinical Excellence (NICE) stated that evidence on partial replacement of the meniscus of the knee using a biodegradable scaffold raised no major safety concerns, but evidence for any advantage of the procedure over standard surgery was limited (NICE, 2012). Therefore, NICE recommends that this procedure should only be used with special arrangements for clinical governance, consent and audit or research.

The American Academy of Orthopaedic Surgeons stated in 2009 that a meniscal transplant may be recommended for active people younger than 55 years-old, with the goal of replacing the meniscus cushion before the articular cartilage is damaged (American Academy of Orthopaedic Surgeons, 2009). The hope is that the transplant will also delay the development of arthritis, but long-term results are not yet available. The website also notes that "synthetic (artificial) meniscal tissue has been tried, but there is conflicting information at this time."

2015 Update

A literature search conducted through February 2015 did not reveal any new information that would prompt a change in the coverage statement. The key identified literature is summarized below.

Collagen Meniscus Implants

Kempshall and colleagues found that functional outcomes following allograft transplantation in knees with greater cartilage damage was similar to knees with lesser cartilage damage when cartilage repair procedures had been employed, although implant survival was lower (Kempshall, 2015).

A retrospective review of 34 patients (17 partial medial meniscectomy and 17 CMI) found no significant difference between the groups for pain and function scores at an average of 9.6 years follow-up (Bulgheroni, 2014).

Polyurethane Meniscal Implant

In contrast with the positive results found in the case series, a controlled pragmatic trial found no benefit of inserting an Actifit at the time of high tibial osteotomy compared to patients who were left with a meniscus defect (Gelber, 2015).

2017 Update

A literature review conducted using the MEDLINE database did not reveal any new literature that would prompt a change in the coverage statement. There were no new randomized controlled trials identified.

2018 Update

A literature search was conducted through November 2018. There was no new information identified that would prompt a change in the coverage statement. The key identified literature is summarized below.

Randomized Controlled Trials

Smith et al reported on the results of a small RCT that randomized 21 patients with a symptomatic meniscal deficient knee to MAT (n=10) or personalized physical therapy (n=11) (Smith, 2018). Another 15 patients who were screened for the RCT decided instead to choose their treatment (referred to as preference group) received MAT (n=6) or personalized physical therapy (n=9). The Knee Injury and Osteoarthritis Outcome Score (KOOS), International Knee Documentation Committee (IKDC) score, Lysholm Knee Scoring Scale score, and complications were collected at baseline, 4 and 8 months, and 1 year after the interventions. Trialists reported pooled results from the RCT and preference group, with statistically significant differences in favor of MAT group for KOOS composite score (mean difference, 12; p=0.03) and KOOS subscales of pain (mean difference, 15; p=0.02) and activities of daily living (mean difference, 18; p=0.005). However, pooling data from the RCT and preference group precluded a meaningful interpretation of data.

Systematic Reviews

Houck et al published the results of a systematic review that included multiple scaffold implantations including CMI (Houck, 2018). No studies in addition to those previously summarized by Warth20 were cited in this systematic review and Houck is not discussed further.

2019 Update

Annual policy review completed with a literature search using the MEDLINE database through November 2019. No new literature was identified that would prompt a change in the coverage statement.

2020 Update

Annual policy review completed with a literature search using the MEDLINE database through November 2020. No new literature was identified that would prompt a change in the coverage statement.

2021 Update

Annual policy review completed with a literature search using the MEDLINE database through November 2021. No new literature was identified that would prompt a change in the coverage statement.

2022 Update

Annual policy review completed with a literature search using the MEDLINE database through November 2022. No new literature was identified that would prompt a change in the coverage statement.

2023 Update

Annual policy review completed with a literature search using the MEDLINE database through November 2023. No new literature was identified that would prompt a change in the coverage statement.

2024 Update

Annual policy review completed with a literature search using the MEDLINE database through April 2024. No new literature was identified that would prompt a change in the coverage statement.

2025 Update

Annual policy review completed with a literature search using the MEDLINE database through April 2025. No new literature was identified that would prompt a change in the coverage statement. The key identified literature is summarized below.

Houck et al and Han et al published results of systematic reviews that included multiple scaffold implantations including CMI (Houck, 2018; Han, 2025). No new studies comparing CMI to meniscectomy were identified in these reviews and they are not further discussed.

CPT/HCPCS:	29868	Arthroscopy, knee, surgical; meniscal transplantation (includes arthrotomy for meniscal insertion), medial or lateral
	G0428	Collagen meniscus implant procedure for filling meniscal defects (e.g., cmi, collagen scaffold, menaflex)

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Group specific policy will supersede this policy when applicable. This policy does not apply to the Wal-Mart Associates Group Health Plan participants.

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